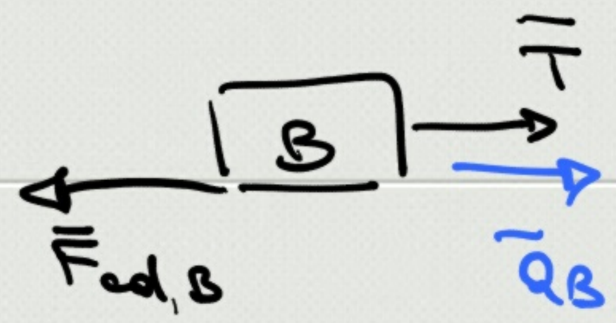
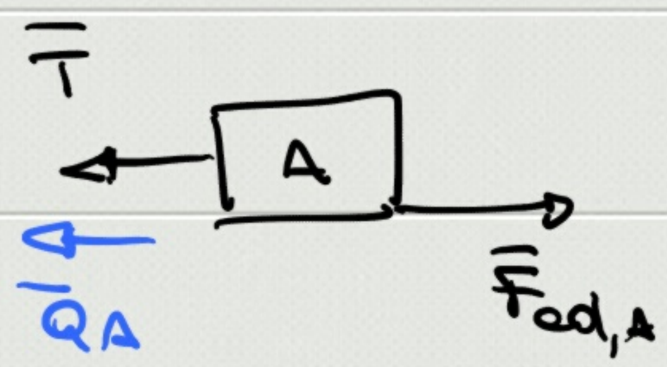
$m_A = 48$ kg

$v_{0A} = v_{0B} = 0$ $\quad \mu = 0.2$ $\quad m_B = 40$ kg

$a_{CM} = ?$





$$\bar{R}^E = m_{TOT}\,\bar{a}_{CM} \quad *$$

$$\bar{a}_{CM} = \frac{m_A \bar{a}_A + m_B \bar{a}_B}{m_A + m_B} \quad *$$

$$\begin{cases} \mu m_A g - T = m_A a_A \\ T - \mu m_B g = m_B a_B \end{cases}$$

$$\cancel{a_{CM} = 0}$$

$$a_{CM} = \cancel{a_A} - a_B$$

$$a_{CM} = \frac{(\mu m_A g \cancel{-T}) + (\cancel{T} - \mu m_B g)}{m_A + m_B} = \frac{m_A - m_B}{m_A + m_B}\mu g = 0.18 \ m/s^2$$

$$\left.\begin{aligned} R^E &= \mu m_A g - \mu m_B g \\ &= (m_A + m_B)\, a_{CM} \end{aligned}\right\} \Rightarrow a_{CM} = \frac{m_A - m_B}{m_A + m_B}\mu g$$

$a_B = 2a_A \qquad T = ?$

$$\begin{cases} \mu m_A g - T = m_A a_A \\ T - \mu m_B g = m_B \cdot 2a_A \end{cases} \qquad \begin{cases} \mu m_A g - T = m_A a_A \\ 2T - \mu m_B g = m_B \cdot 2a_A \end{cases}$$

$$\begin{cases} \mu m_A g - 2T = m_A a_A \\ T - \mu m_B g = -m_B \cdot 2a_A \end{cases} \qquad \begin{cases} \mu m_A g - T = m_A a_A \\ T - \mu m_B g = -m_B \cdot 2a_A \end{cases} \quad *$$

$$\mu g (m_A - m_B) = a_A (m_A - 2m_B)$$

$$\Rightarrow a_A = \mu g \frac{m_A - m_B}{m_A - 2m_B}$$

$$\Rightarrow T = \mu g m_A \left(1 - \frac{m_A - m_B}{m_A - 2m_B}\right) =$$

$$= \mu g m_A \frac{\cancel{m_A} - 2m_B - \cancel{m_A} + m_B}{m_A - 2m_B} = \mu g m_A \frac{m_B}{2m_B - m_A} \simeq$$

$$= 118 \text{ N}$$